

Numericals:

$$X = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

$$d=3$$

reduce the dimensionality & compute the variance when

$$(i) d=3 \rightarrow d=2$$

$$(ii) d=3 \rightarrow d=1$$

Step 1: mean centering

$$\mu = [1.5 \quad 1.5 \quad 1.5]$$

(mean)

$$X_c = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix}$$

Step 2: Compute covariance matrix

$$C = \frac{1}{n} X_c^T X_c$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ x_1 & \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix} \\ x_2 & \\ x_3 & \end{matrix}$$

$$\begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) & \text{cov}(x_1, x_3) \\ \text{cov}(x_2, x_1) & \text{cov}(x_2, x_2) & \text{cov}(x_2, x_3) \\ \text{cov}(x_3, x_1) & \text{cov}(x_3, x_2) & \text{cov}(x_3, x_3) \end{bmatrix}$$

Step 3: Eigen values & Eigen vectors

$$\lambda_1 = 2.0, \quad \lambda_2 \approx 0.75, \quad \lambda_3 \approx 0.0$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Step 4: Dimensionality reduction.

$$3D \longrightarrow 2D$$

Take top 2 eigen vectors

$$W_{2D} = [v_1 \quad v_2]$$

Project the data points

$$Z = X_c W_{2D}$$

$$= []$$

Each point will be of 2D

$$3D \longrightarrow 1D$$

Take top 1 eigen vector

$$W_{1D} = [v_1]$$

Project the data point

$$Z = X_c W_{1D}$$

$$\text{Total variance} = 2 + 0.75 + 0$$

$$= 2.75$$

$$PC_3 = 0\%$$

$$PC_1 = \frac{2}{2.75} \approx 73\%$$

$$PC_2 = \frac{0.75}{2.75} \approx 21\%$$